

Name: _____

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Class: _____

MyDesign Portfolio

Element K Initial Template

Element K: Reflection on the Design Project

Engineering Design Process Summary (K1, K2)

Summarize each step of the design process including:

- ~~defining the problem (elements A-C)~~
- ~~ideation (element D)~~
- ~~selection & prototype design (elements D-F)~~
- prototype building (element G)
- testing (elements H-I)
- ~~evaluation (element J)~~

For each project step, explain: What went well? What didn't go well? Why?

Step 1: Figuring out the problem

- What is the problem?
- What caused the problem?
- How can we fix the problem?
- How big is the problem?

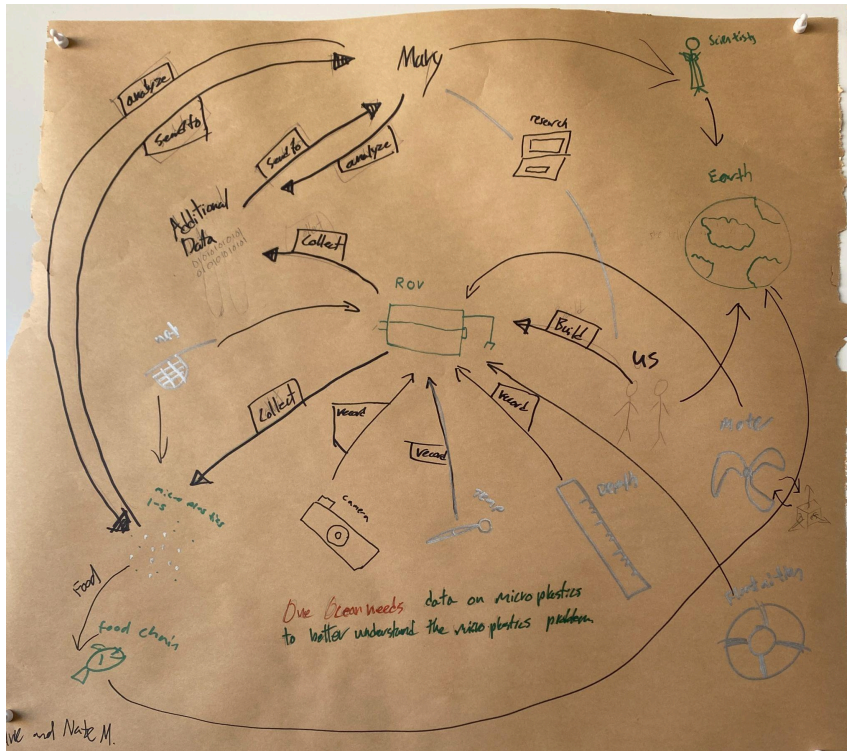
We then wrote a problem statement from the answers to these questions and more. This went well because we could define the issue we needed to fix.

Step 2: Figuring out what our ROV needs to solve this problem

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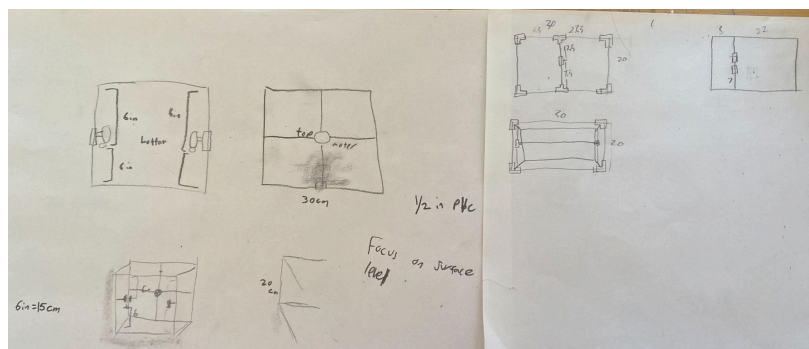
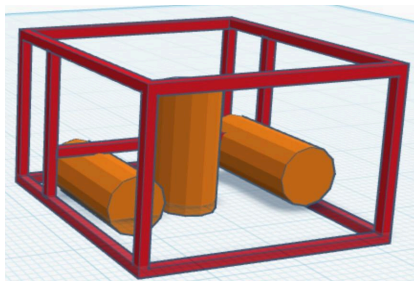
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Defining why we are building an ROV that collects microplastics. "One Ocean needs data on microplastics to better understand how they affect the world." - Elvie. This went well because it did what we needed to do, give us the reason we were building this robot.

Step 3: Deciding on size and motor placement for ROV

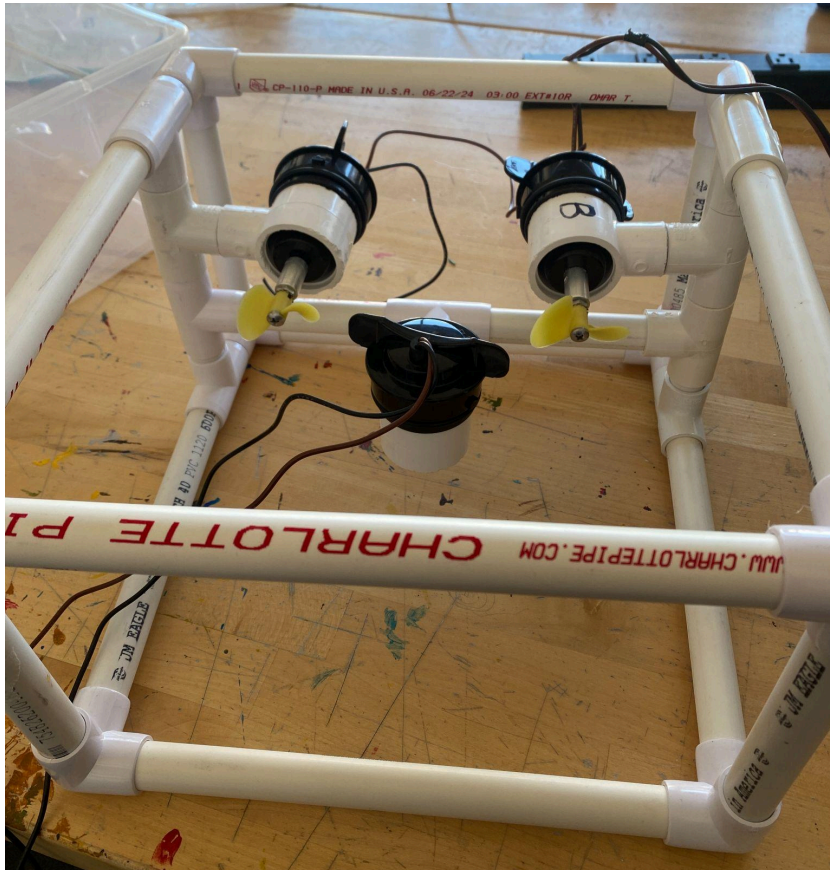


This did not go very well because it was hard for us to visualize what the other was picturing. In the end, Nate created a simple model in TinkerCAD which we both agreed on.

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Step 4: Building the frame and attaching the motors

This went well. The frame was built quickly, and the motors were attached. We did not change the shape of the frame after this point. We only added to it.

Step 5: Creating Task List

We brainstormed what small tasks your group needs to do to complete the creation of the ROV on time. This went well because we thought of many things that needed to be done. Some of these tasks did not happen, like adding a camera, because we did not have the time or resources to complete them.

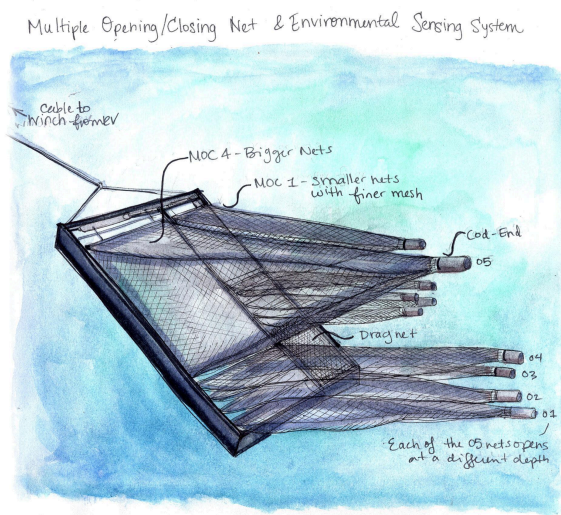
Step 6: Researching other MP devices

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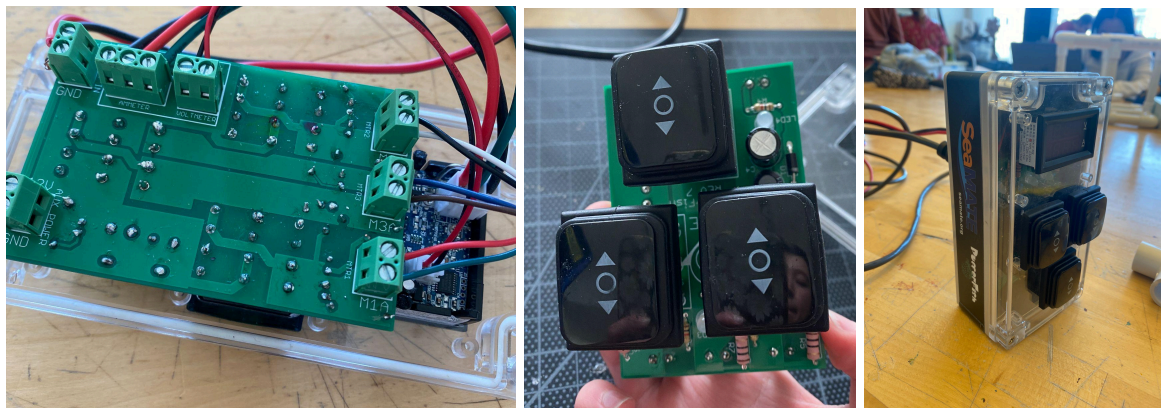
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We did research on other MP devices and how they worked.



This went well because it helped us brainstorm what we wanted our MP collection device to look like.

Step 7: making the puffer fish remote



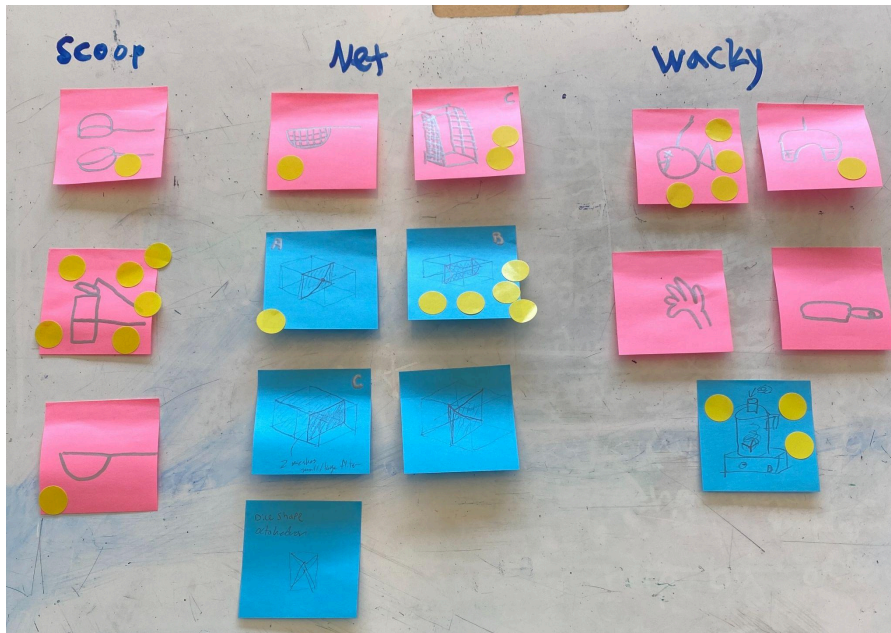
What went well was that we both improved our soldering technique, and we finished creating it before the other groups. We built this by following the slideshow and reading their steps carefully.

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Step 8: Brainstorming for the MP collection device



We thought of many practical and fun and unique ways to collect MP, which went very well. Then the whole class voted on which ideas they liked the most.

Step 9: Weighted Decision Matrix For Micro Plastic Collection

Criteria	weight	Net	Test Tube
Sample Sizes	0.4	1.0	0.3
Integration *	0.15	0.6	0.45
Reusability	0.2	0.4	0.7
Ease of Collection	0.15	0.4	0.7
Affordability	0.07	0.55	0.3
Uniqueness	0.03	0.3	0.4
score		0.68	0.47

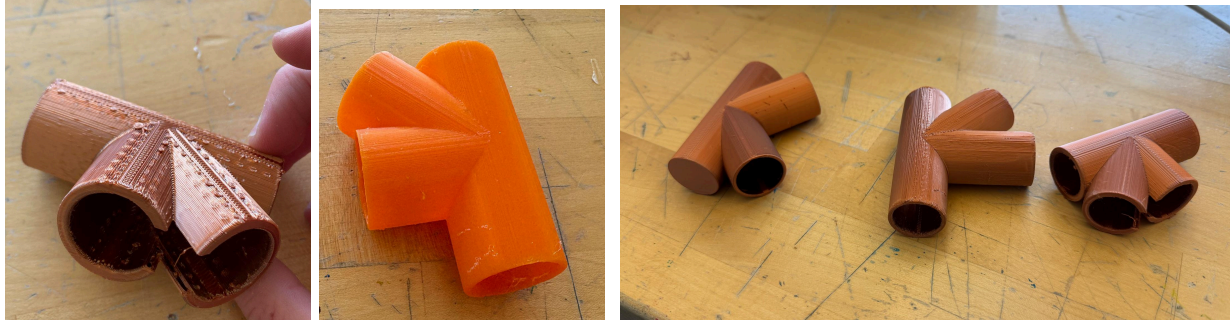
* how well it attaches to ROV

We listed the criteria we thought was important for the MP collection device to have, then we rated each of our ideas based on that. This helped us decide which criteria was more important and by how much.

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Step 10: Connections - frame and microplastic device

This model was created and adjusted on CAD. There were many versions of this collection device connections that were printed and did not work. This stage was our biggest struggle because even the ones that fit were very hard to insert all the PVC pipes in to create a triangle shape. Every single one of them end up withstanding a crack by the end of the build process.

Step 11: Making the shape for the MP device

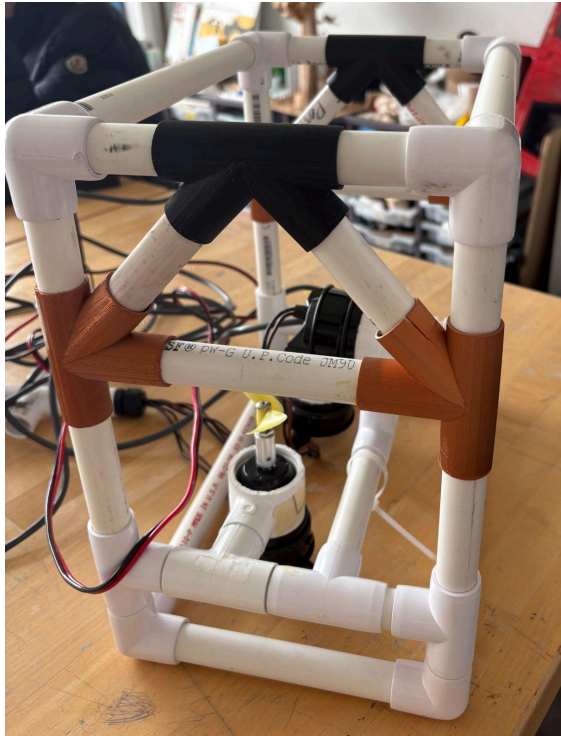
I cut and moved around PVC so they all fit together attached to the frame in a pyramid shape.



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This did not go well because there were no measurements accessible, and the connections broke a bit when being put together.

Step 12: Mesh

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- Cut and hot glued big three faces
 - Top of triangle big mesh
 - Bottom thin mesh
- Attached permanently one side with big mesh
- Put snaps and side pieces to make flush the other side so it could be removed and MP's could be retrieved
- Reinforced where all the loose hot glue was with fishing wire and sewing needle

At first this did not go, but with the quick problem-solving it turned out better than what we expected.

Step 13: Buoyancy

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Pool floaties were added and subtracted from the frame to figure out how to make the ROV neutrally buoyant. This was tested in a bucket and after was secured with zip ties. This went well because it was very simple just to guess and check for the pool floaties instead of doing the math to figure this out.

Step 13: Testing at Drew

Both shake test and buoyancy were tested at Drew.

The shake test was shaking the robot and went well because everything stayed together. This test showed that we needed to add drainage to the ROV. Many holes were drilled into the frame, so after a test, water would drain out of the frame and become easier to transport.

Buoyancy was tested by adding and subtracting pool floaties to see how the ROV would react. This test went well because it helped us figure out how many floaties we needed to achieve the buoyancy we wanted.

Step 14: Testing at JCC

Testing at JCC helped us nail down the buoyancy to make it closer to neutral rather than just buoyant, allowing us to dive when needed. This test showed that our controls were responsive and intuitive, allowing us to easily maneuver through the pool. There were no

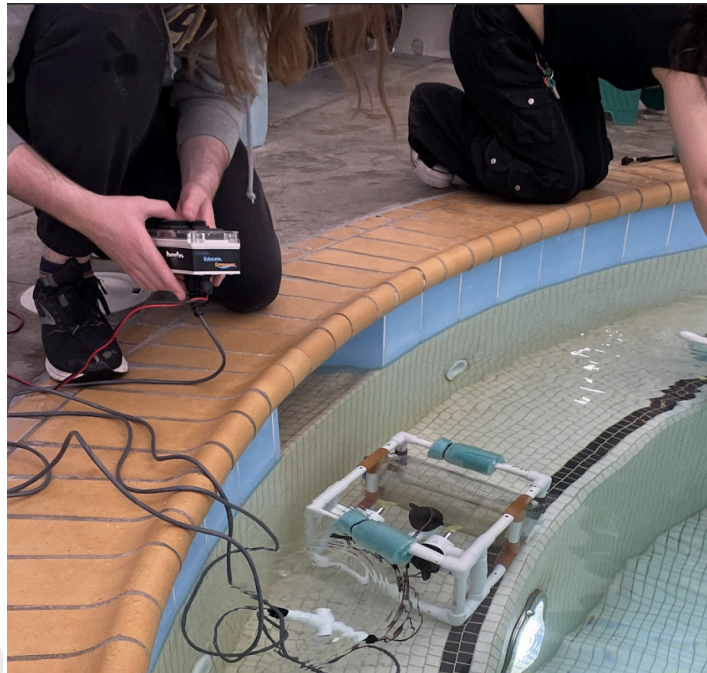


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problems maneuvering the ROV to any possible position and the entire ROV stayed more or less upright.



 IMG_4502.MOV

^^Video

Step 15: Testing at Fort Mason

We put the ROV in the bay and drove it around. This successfully collected some MP that were in the water, and the robot smoothly maneuvered through the water. Additionally we reaffirmed our conclusion from the JCC, our ROV maneuvered really well.

 ROV testing at Fort Mason (Elvie and Nate)

Lessons Learned About the Engineering Design Process (K3)

What do you think designers need to consider or do based on your experience? A level 5 reflection illustrates thorough and specific details on improving the design process in the future. Make sure to touch on lessons from across the whole engineering design process and experience that can apply to any future engineer. The lessons learned in Element K

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should focus on the engineering design process and how you implemented it. Later on (in L1) you will provide recommendations for continuing work on this project in particular.

The main recommendations around creating the ROV that are harder to document is that the buoyancy needs to be adjusted a lot. Not only do you need to aim for neutral buoyancy, but you want the ROV to remain upright, which means balancing out the density across the ROV. We let our floaties slide a little bit to adjust everything, but we found that the positions had to be readjusted everytime we drove the ROV.

Another thing is that it's helpful to have the ROV drain and fill up with water fast. This means adding holes where you notice that there are still bubbles coming from after a few seconds in the water.

